

Syeda Jannatus Saba

sysaba@cs.stonybrook.edu | Stony Brook, NY | LinkedIn | Google Scholar | GitHub

EDUCATION

- **Stony Brook University, New York, USA** Aug 2023 – Present
Ph.D. in Computer Science | Research Area: Machine Translation & Evaluation | GPA: 3.95 / 4.00
- **Shahjalal University of Science and Technology, Sylhet, Bangladesh** Feb 2017 - Sep 2021
B.Sc. in Computer Science and Engineering | GPA: 3.96/4.00

EXPERIENCE

- **Stony Brook University, NY** Jun 2024 - Present
Graduate Research Assistant Advisor: Steven Skiena
 - Developed a **synthetic data generation framework** for training translation-evaluation models via iterative “telephone” translations, achieving state-of-the-art correlation with human judgments and surpassing xCOMET (EMNLP 2025, Oral).
 - Investigated **constrained text generation** with LLMs by producing an ‘e’-less translation of *The Great Gatsby*; analyzing semantic fidelity under increasing letter restrictions (IJCNLP-AACL 2025, Oral).
 - Building a **multilingual textbook translation and PDF-to-LaTeX pipeline**, ensuring book-level consistency, glossary accuracy, and stylistic preservation across languages.
- **SafeBeat Rx Inc, San Francisco, CA (YC S21)** Nov 2022 - Aug 2023
Software Engineer (Remote, Contract) Tech Lead: Alejandro Alvarez Martinez
 - Optimized graph visualization pipelines with efficient data structures to improve rendering on large datasets.
 - Implemented **interactive graph exploration** algorithms (zoom, pan, spatial layout) on canvas-based systems.
 - Built component library from scratch for new design system to rewrite legacy frontend codebase.
- **Re:frame LLC, Wilmington, DE** Dec 2021 – Nov 2022
Software Engineer (Remote, Contract) Tech Lead: Dhananjoy Biswas
 - Built a Visitor Pattern-based system for abstract syntax trees to automate code validation and refactoring.
 - Integrated Prisma, Planetscale, and tRPC for scalable backend communication.

SELECTED PUBLICATIONS

Evaluating Language Translation Models by Playing Telephone [paper]

Syeda Jannatus Saba, Steven Skiena

EMNLP 2025 (ORAL)

A Study on Using Semantic Word Associations to Predict the Success of a Novel [paper]

Syeda Jannatus Saba, Biddut Sarker Bijoy*, Henry Gorelick, Sabir Ismail, Md Saiful Islam, Mohammad Ruhul Amin*

*SEM 2021 (CO-LOCATED WITH ACL-IJCNLP 2021)

Gatsby without the ‘E’: Creating Lipograms with LLMs [paper]

Rohan Balasubramanian, Nitish Gokulakrishnan, Syeda Jannatus Saba, Steven Skiena

IJCNLP-AACL 2025 (ORAL)

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, vLLM, Instruction Tuning, Distributed Training (DDP)
- **Natural Language Processing:** Hugging Face, spaCy, PEFT, LoRA/qLoRA, Prompt Engineering
- **Tools:** Python, C++, SQL, Pandas, NumPy, CUDA, Git, Docker, Linux, Weights & Biases, Bash, SLURM, Jupyter

HONORS AND PROFESSIONAL ACTIVITIES

- University **Dean’s Award** (2023): Highest undergraduate GPA in the School of Applied Sciences & Technology.
- **Program Committee & Reviewer**, Bangla Language Processing (BLP) Workshop @ IJCNLP–AACL 2025.
- ACM ICPC Dhaka Regional Programming Contest (2019): **Top 15%** among 300+ teams.
- National Girls’ Programming Contest (2018, 2019): **Top 5** rank among 100+ teams.
- **CodeForces:** Max Rating 1774 ([Valkyrie28](#)), Top 1.1% **CodeChef:** Max Rating 2124 ([kyrieo.0](#)), Top 0.5%.
- Bangladesh Regional Mathematical Olympiad (2015): Secured **2nd Runner-up** position.